

Question Bank for InSem 1

CO 1

1. Outline the concept of scientific computing and illustrate it with one application area.
2. Interpret computational cluster in High-Performance Computing (HPC) and state its purpose.
3. Infer any two commonly used performance metrics in scientific applications.
4. Illustrate the basic components of an HPC architecture with a neat explanation.
5. Explain the concept of scalability in HPC and distinguish between strong and weak scaling.
6. Infer the architecture of a cluster-based HPC system, including compute nodes, interconnects, and storage.
7. Extend the key performance evaluation metrics used to assess scientific computing applications.
8. Illustrate numerical computation and its purpose in scientific computing.
9. Outline workflows in scientific computing and their significance.
10. Interpret the concept of speedup in the context of performance measurement.
11. Explain the importance of parallelism in HPC systems.
12. Rephrase the purpose of simulation pipelines in scientific experiments.
13. Demonstrate the different types of performance metrics and their significance in evaluating HPC workloads.
14. Explain the role of a compute node in an HPC cluster.
15. Interpret efficiency in the context of parallel computing.
16. Extend the need for scalable architectures in scientific computing.
17. Outline the basic structure of a scientific workflow.
18. Demonstrate the scalability challenges in large-scale scientific computing environments.
19. Interpret the importance of performance measurement in scientific simulations.
20. Infer a simulation pipeline in detail, highlighting each stage involved.
21. Summarize the concept of scientific computing.
22. Interpret the concept of a computational cluster.
23. Outline the basic architecture of a High-Performance Computing (HPC) system.
24. Relate the concept of scalability in scientific computing, highlighting its importance.
25. Summarize cluster-based HPC architecture with suitable components and their functions.
26. Illustrate the stages involved in a scientific computing workflow.
27. Interpret different performance metrics used to evaluate scientific computing applications.
28. Summarize the concept of numerical computation.
29. Outline the stages involved in a simulation pipeline.
30. Illustrate the role of clusters in large-scale scientific problem solving.
31. Compare the strong scaling and weak scaling.
32. Interpret the end-to-end process of scientific simulation, from modeling to result analysis.
33. Show the importance of simulation pipelines in scientific experiments.

34. Relate the significance of performance evaluation in HPC-based scientific applications.
35. Summarize any two objectives of scientific computing.
36. Interpret the role of a compute node in a computing system.
37. Explain the need for High-Performance Computing in scientific applications.
38. Show the components of an HPC cluster and their roles.
39. Extend the workflow model used in scientific computing environments.
40. Interpret how numerical methods support scientific computations.
41. Demonstrate various metrics used to measure scalability and performance in HPC systems
42. Summarize the concept of performance metrics in scientific computing.
43. Interpret the concept of parallel computing.
44. Summarize the architecture and working of a scientific computing cluster.
45. Demonstrate the importance of workflows in managing scientific applications.
46. Outline the design and execution of simulation pipelines in scientific computing.
47. Infer the relationship between problem size and scalability.
48. Rephrase how HPC architectures enhance computational performance for large-scale scientific problems.

CO 2

1. Summarize the concept of cloud computing and mention one benefit it provides for scientific workloads.
2. Classify any two scientific application domains that require high computational power.
3. Demonstrate GPU is and its role in hardware acceleration.
4. Summarize the role of virtual machines and hypervisors in cloud computing environments.
5. Demonstrate how scientific computing is applied in climate modeling.
6. Apply the concept of GPU acceleration to improve performance in an engineering simulation.
7. Make use of heterogeneous computing principles along with CPU and GPU for a physics-based simulation problem.
8. Classify any two hardware accelerators used in scientific computing.
9. Summarize one scientific application of computing in the field of biology.
10. Show the use of scientific computing in engineering design and analysis.
11. Apply containerization concepts to ensure portability of scientific software across cloud platforms.
12. Classify heterogeneous computing based on the use of multiple processing units.
13. Explain the differences between public, private, and hybrid cloud models.
14. Develop a cloud-based deployment strategy using Docker and Kubernetes for an engineering application.

15. Demonstrate the use of GPU vs CPU for accelerating a numerical computation task.
16. Utilize hardware acceleration techniques to optimize performance in multidisciplinary scientific workloads.
17. Illustrate how scientific computing supports engineering applications.
18. Outline the role of virtual machines in cloud computing.
19. Illustrate the layered cloud architecture used for scientific workloads.
20. Apply containerization using Docker to deploy a scientific application in the cloud.
21. Build a Kubernetes-based deployment for scalable execution of a scientific workload.
22. Identify how GPU acceleration can improve performance in a physics- based simulation.
23. Make use of heterogeneous computing principles using CPU and GPU for an engineering problem.
24. Illustrate the importance of cloud computing in climate modeling.
25. Interpret the function of a hypervisor in a virtualized environment.
26. Extend public, private, and hybrid cloud models for scientific workloads.
27. Apply Kubernetes orchestration for managing containerized scientific applications.
28. Develop a hybrid cloud strategy for large-scale climate simulation workloads.
29. Identify the use of TPUs for accelerating scientific data processing tasks.
30. Utilize FPGA-based acceleration to optimize a computation-intensive scientific task.
31. Illustrate the role of scientific computing in biological data analysis.
32. Outline the concept of containerization.
33. Illustrate the working of Docker and Kubernetes in cloud-based scientific applications.
34. Apply virtual machine provisioning to execute a scientific simulation in the cloud.
35. Construct a cloud-based execution model for biological simulations using containers.
36. Identify how GPU and CPU coordination improves application performance.
37. Make use of heterogeneous hardware acceleration for multidisciplinary scientific workloads.
38. Illustrate the role of scientific computing in physics research.
39. Compare CPU and GPU architectures.
40. Extend the design and role of cloud infrastructure layers in scientific computing.
41. Apply container-based deployment for a scientific application requiring scalability.
42. Construct an end-to-end cloud solution for engineering simulations using Docker and Kubernetes.
43. Develop the application of TPU acceleration for machine-learning-driven scientific workloads.
44. Make use of heterogeneous computing with CPU, GPU, and FPGA for optimized scientific execution.